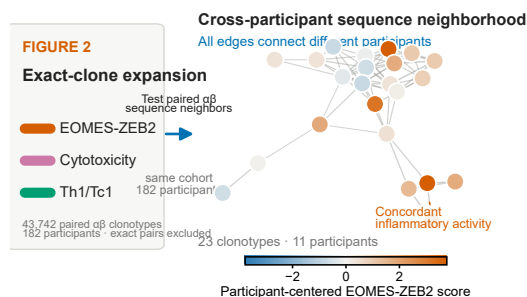
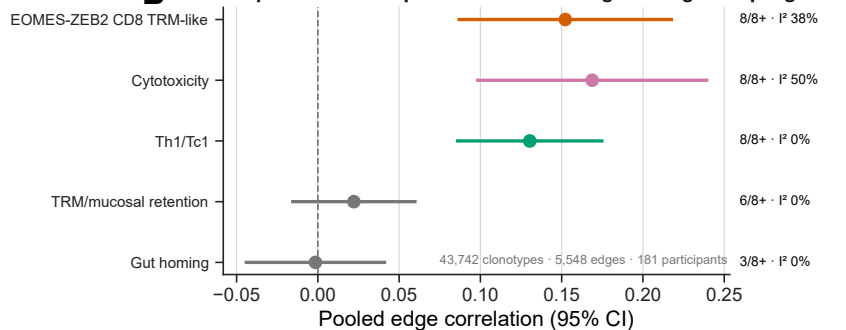


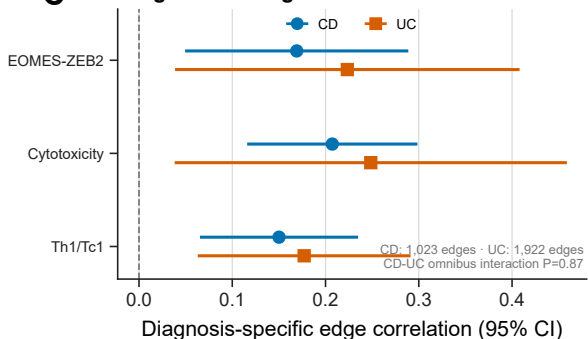
A Expansion-linked programs recur in $\alpha\beta$ neighborhoods



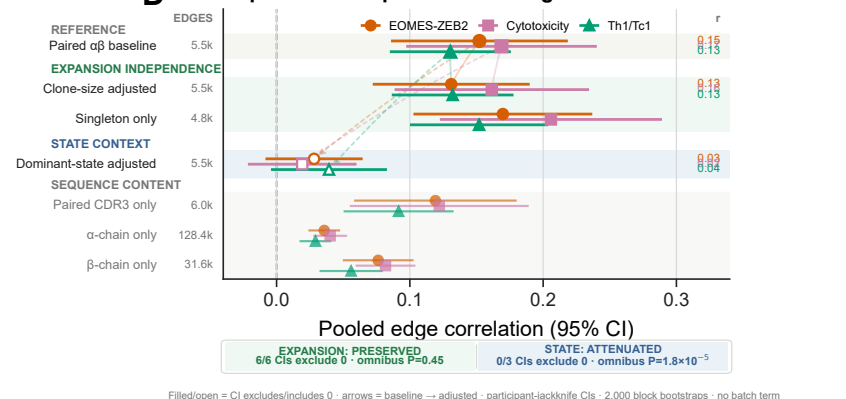
B Sequence-related paired TCRs converge on Figure 2 programs



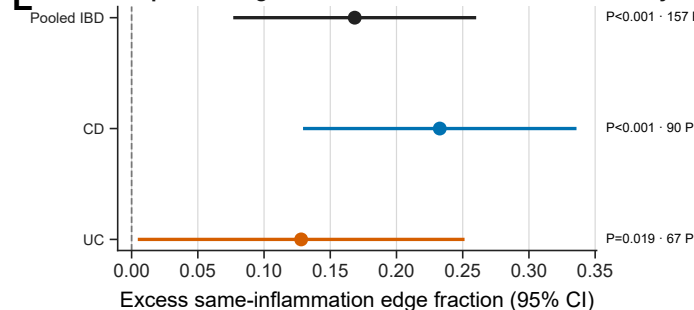
C Program convergence is conserved in CD and UC



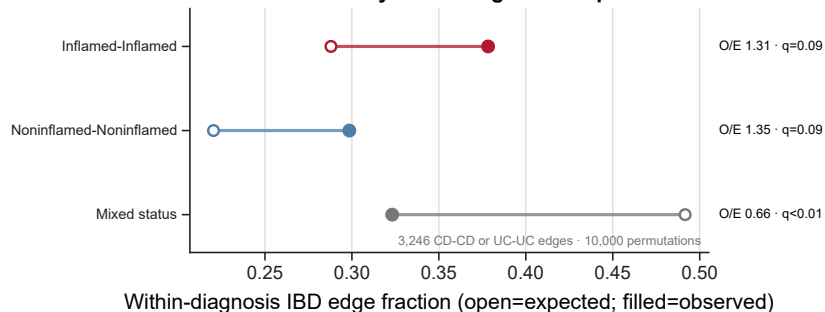
D Expansion-independent convergence is state-sensitive



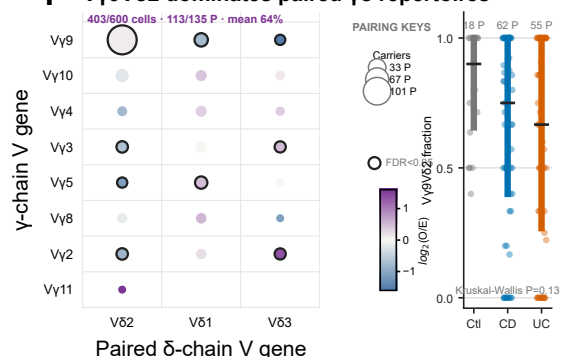
E Sequence neighborhoods associate with inflammatory status



F Mixed inflammatory-status edges are depleted



F V γ 9V δ 2 dominates paired $\gamma\delta$ repertoires



G Expansion reveals $\gamma\delta$ effector shifts beyond V γ 9V δ 2 identity

